Order amidst a world of chaos

Recent developments in cluster synchronization

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Motivation

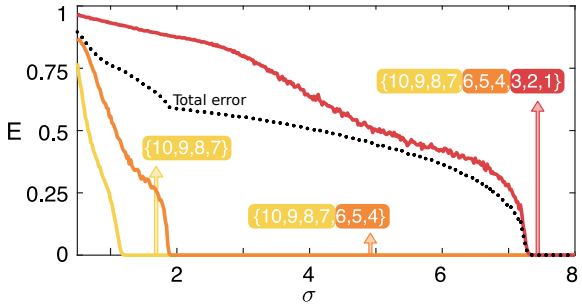
- Synchronization is to this day, a phenomena which captures the imagination of scientists of different disciplines.
- Perplexing as it might appear, synchronization can also be observed within populations of chaotic units, often at high couplings.
- Does the transition to such high couplings feature some substructures before the network is completely in sync?

2. Intermediate transitions and clusters

- A witness for synchronization is the synchronization error E_k: the standard deviation of the states $\mathbf{x}_i$ of nodes within a group C_k at late times $T \gg 1$,

$$E_k = \left\langle \sqrt{\frac{1}{N_k} \sum_{i \in C_k} |\mathbf{x}_i - \overline{\mathbf{x}}_k|^2} \right\rangle_T .$$

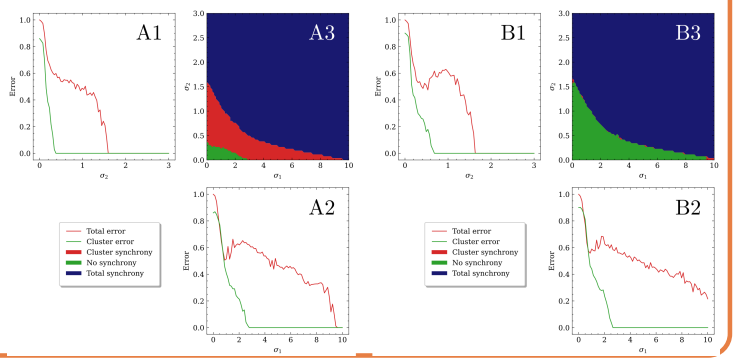
- In all kinds of networks, the synchronization error tells an intriguing story: before the coupling is strong enough that the entire network is in sync, there are already groups of nodes (clusters) displaying synchronous dynamics.



- How do we tell them apart? At what values of the coupling do these groups synchronize?

4. A deeper dive: equitability

- In [3] we achieved a better understanding of the necessary conditions for clusters to form, based on the novel concept of "dynamical equitability".
- This allows us to understand more general situations than those described with localized spectral blocks.
- It also explains why cluster synchronization is evasive in multilayer networks and hypergraphs. In the former we showcase this comparing two similar networks which synchronize on each layer separately, but only one also achieves cluster synchronization when both layers are active.

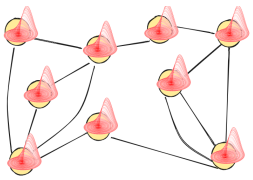


1. Standard tools in synchronized chaos

- We work under two assumptions, standard in the literature: identical units and diffusive coupling,

$$\dot{\mathbf{x}}_i = \mathbf{f}(\mathbf{x}_i) + \sigma \sum_{j=1}^{N} \mathcal{L}_{ij}\, \mathbf{g}(\mathbf{x}_j),$$

where $\mathbf{f}(\mathbf{x})$ is the node's internal dynamics (chaotic, e.g. Lorenz or Rössler) and $\mathcal{L}$ is the graph Laplacian.



- The Master Stability Function framework [1] is a well established procedure to assess the conditions under which synchronization is expected.
- In a nutshell, one computes the Maximum Lyapunov Exponent (MLEs) associated to perturbations $\boldsymbol{\eta}_i$, one per Laplacian eigenpair $(\lambda_i, \mathbf{v}_i)$,

$$\dot{\boldsymbol{\eta}}_i = [J_f(\mathbf{x}_s) + \sigma \lambda_i J_g(\mathbf{x}_s)]\boldsymbol{\eta}_i,$$

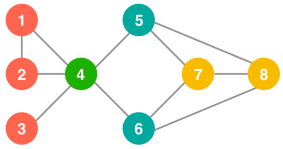
where $J_f(\mathbf{x})$, $J_g(\mathbf{x})$ are Jacobians, and $\mathbf{x}_1(t) = \cdots = \mathbf{x}_N(t) \equiv \mathbf{x}_s(t)$ is the synchronous solution.
- Synchronization requires all MLEs to be negative (except for the one from $\lambda_1 = 0$, which by definition is positive), i.e. perturbations must be damped.

3. Spectral blocks and where to find them

- Clusters synchronizing coincides with certain eigenvalue transitions: all perturbations affecting a given set of nodes are damped.
- Which perturbations (dictated by the Laplacian eigenvectors) affect which nodes? We show in [2] the following:

Theorem 1. *Localized spectral block $\Longrightarrow$ synchronization cluster.*

where a localized spectral block is a group of nodes satisfying certain conditions related to the Laplacian eigenvectors.
- Localized spectral blocks are intrinsically related to an older concept in graph theory: external equitable partitions, i.e., groups of nodes receiving equal external input.



- We also provide an algorithmic method to find these localized spectral blocks, and the coupling values at which the corresponding nodes synchronize, exploiting the properties of the Laplacian eigenvectors.

Summary

- We provided an analytical description of the different cluster-synchronized stages in generic networks of chaotic oscillators, which is independent of the specific dynamics under consideration.
- We designed a algorithm to detect the transitions, and exemplified it with different real networks.
- We extended traditional notions of equitability to handle unaccounted phenomena and applied it to multilayers and hypergraphs.

KEY REFERENCES

[1] L. M. Pecora, T. L. Carroll. "Master Stability Functions for Synchronized Coupled Systems". *Physical Review Letters 80* (1998).

[2] A. Bayani, F. Nazarimehr, S. Jafari, K. Kovalenko, G. Contreras-Aso, K. Alfaro-Bittner, R. J. Sánchez-García, S. Boccaletti. "The transition to synchronization of networked systems". *Nature Communications 15, 4955* (2024).

[3] K. Kovalenko, G. Contreras-Aso, C. I. del Genio, S. Boccaletti, R. J. Sánchez-García. "Equitability and explosive synchronisation in multilayer and higher order networks". *arXiv:2507.09319*.

MORE INFORMATION

Nat. Comm. [2]

Comm. Phys. [3]

Funding: Financiado por la Unión Europea NextGenerationEU